

Demonstration of Proposed Features

Date	2 July 2026
Team ID	XXXXXX
Project Name	Rising Waters
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Authentication	Email/password signup with verification, Google OAuth login, password reset, and login history	Implemented	Yes	Includes OTP-based email change from profile
2	Profile Management	Manage username, phone number, and profile picture (via Google); email change with OTP confirmation	Implemented	Yes	
3	Flood Prediction Engine	ML-based inference using 5 features: annual rainfall, cloud visibility, temperature, humidity, and seasonal rainfall	Implemented	Yes	Core prediction feature of the app
4	ML Model Registry	4 trained models stored in the database: Random Forest, Gradient Boosting, Logistic Regression, and XGBoost	Implemented	Yes	Models trained via seed_models.py
5	Dashboard	Aggregated stats, flood risk distribution, and a 7-day prediction trend chart	Implemented	Yes	
6	Prediction History	Paginated list of past predictions with a per-prediction detail view and text export	Implemented	Yes	
7	Location Monitoring	Save named locations and view live weather plus a 3-day forecast via the Open-Meteo API	Implemented	Yes	No API key required for weather data

8	Automated Weather Alerts	Background job checks saved locations hourly and emails severe-weather warnings	Implemented	Yes	Built with APScheduler; configurable interval
9	AI Flood Bot Chatbot	Gemini 2.5 Flash powered assistant for flood safety and platform questions	Implemented	Yes	

Feature Implementation Summary:

Total Features Proposed	9
Total Features Implemented	9
Total Features Demonstrated	9
Overall Implementation Rate (%)	100%